Aaradhaya Sutrave

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EDUCATION

Texas A&M University (TAMU) | Expected Dec. 2026

- Bachelor of Science in Mechanical Engineering | College Station, TX
- Relevant coursework: Heat Transfer, Dynamics Systems and Controls, Thermodynamics, Fluid Mechanics, Materials and Manufacturing in Design, Computer-Aided Engineering

TECHNICAL SKILLS

Programming & Data Acquisition: MATLAB, Python, LabVIEW, Arduino Programming, Excel

Instrumentation & Electronics: Lock-in Amplifier, Oscilloscope, Function Generator, NSOM probe, circuit simulation

Design & Analysis: SolidWorks, ANSYS, Fusion 360, Orca Slicer, 3D printing

EXPERIENCE

Undergraduate Research Assistant – Sy Bor Wen, Department of Mechanical Engineering May 2026 - Present

- Developed a custom LabVIEW VI for buffered and continuous data acquisition from a lock-in amplifier
- Configured lock-in amplifier sensitivity and low-pass filter settings to extract weak optical signals
- Worked with a function generator, oscilloscope, and NSOM probe to generate and measure optical signals near a microheater
- Contributing to an ongoing research paper on Near Field Radiation

Undergraduate Research Assistant – MIDL Research Lab Feb. 2026 - Present

- Developed MATLAB code to generate Voronoi diagrams and half edge mesh data structures that replicate Finite element analysis mesh
- Designed precision acoustic test apparatus (Kundt tube) parts using SolidWorks CAD; creating PVC tube flanges, speaker reducers and measurement ports. Then converted designs to G-code for 3D printing
- Contributing to an ongoing research manuscript on Noise-absorption of weave structures.

ENGINEERING PROJECTS

CNC Machining and CAM Project – Fusion 360 Apr. 2026

- Created a CAD model, generated CAM toolpaths and tool selection in Fusion 360, and machined a plastic part on a CNC mill

Backseat Minder – Sensor, Circuit, & Alert-System Design Jan. 2026

- Created circuit diagram and simulation for a self-charging monitoring concept with occupant, temperature, and humidity detection
- Designed system functionality for alerts through a cellular module and supported CAD work for PCB integration considerations

Aluminum Hammer Fabrication: Multi Process Manufacturing Nov. 2025

- Fabricated an aluminum hammer using manual milling, lathe turning and precision sanding in 4 - 5 weeks

Aerodynamic Car Design & Wind Tunnel Testing – SolidWorks, Flow Simulation Nov. 2025

- Led a four person team to design and test a low drag car body using SolidWorks and Flow Simulations.
- Validated the design through wind-tunnel testing and achieved the lowest drag coefficients of the 12 teams.

Stepped Shaft FEA/Discontinuity Effects Nov. 2025 - Present

- Conducted FEA of load bearing stepped shaft under different types of loading and constraints using SolidWorks, evaluated bending, von Mises, and shear stress to assess the effects of geometric discontinuities.
- Developed numerical simulations to validate FEA results and investigated effects of geometry and stress concentrations on potential failure spots.

Mars Rover Simulation – Python Modeling & Optimization Oct. 2024

- With a team, built a Python simulation that parameterized rover mass, material properties, and DC motor behavior across randomized Mars-like terrain.
- Evaluated and optimized rover design parameters for environmental constraints

Heat Sink Manufacturing Design – CNC Process Planning Mar. 2024

- Developed a CNC process pan for a machined heat sink, selecting cutting tools, machining parameters, operations, and toolpaths based on part geometry. Tolerances, and material removal requirements.
- Estimated Machining cycle times, production capacity, and machine requirements for manufacturing efficiency, tool life and accuracy.